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ISRO takes off

ISRO launches its first satellite of the year for Brazil. It will be used to monitor deforestation in Brazil. <http://dgit.in/mar21-44>

had to be obtained by scanning point-by-point. The greatest technological problems, he predicted, will be the illumination and detector movement.

ELECTRON MICROSCOPES

An electron microscope is a microscope that illuminates with a beam of accelerated electrons. Electron microscopes have a higher resolving power than light microscopes and can show the structure of smaller structures since the wavelength of an electron is 100,000 times shorter than that of visible light photons. There are multiple types of Electron Microscopes which were invented right after each other. These were:

- Transmission electron microscope (TEM)
- Scanning electron microscope (SEM)
- Reflection electron microscope (REM)
- Scanning transmission electron microscope (STEM)
- Scanning tunneling microscopy (STM)

While all notable updates, the next major milestone was Confocal microscopy. It is an optical imaging technique that uses a spatial pinhole to block out-of-focus light in image formation to increase optical resolution and contrast of a micrograph. Multiple two-dimensional images taken at different depths in a sample allow for the reconstruction of three-dimensional structures within an object (a process known as optical sectioning). Life sciences, semiconductor inspection, and materials science are some of the popular applications of this technique in the scientific and industrial communities.

Marvin Minsky invented the idea of confocal imaging in 1957, with the intention of solving some of the drawbacks of conventional wide-field fluorescence microscopes. The entire specimen is flooded uniformly in light from a light source in a typical fluorescence microscope. The fluorescence generated by all parts of the sample can be excited at the same time and detected by the microscope's photode-

tector or camera, which includes a large unfocused background portion.

A confocal microscope, on the other hand, removes out-of-focus signal by using point illumination and a pinhole in an optically conjugate plane in front of the detector (the term "confocal" comes from this configuration). The image's optical resolution, especially in the sample depth direction, is much better than that of wide-field microscopes because only light emitted by fluorescence very close to the focal plane can be detected. However, since most of the light from the sample fluorescence is blocked at the pinhole, the improved resolution comes at the expense of signal amplitude, necessitating long exposures. To compensate for the signal drop after the pinhole, a sensitive detector, such as a photomultiplier tube (PMT) or an avalanche photodiode, measures the light intensity and converts it to an electrical signal.

STM

STM stands for scanning tunneling microscope and is used to image surfaces at the atomic level. In 1986, its inventors, Gerd Binnig and Heinrich Rohrer, then at IBM Zürich, were awarded the Nobel Prize in Physics for their work in 1981. STM detects surface features with a 0.01 nm depth resolution by using an exceptionally sharp conducting tip that can discern features smaller than 0.1 nm. Individual atoms can now be imaged and manipulated on a daily basis. The majority of microscopes are designed for use in ultra-high vacuum at temperatures reaching zero kelvin, but there are some that can be used in air, water, and other settings, as well as at temperatures exceeding 1000 °C.

Quantum tunnelling is the cornerstone of STM. A bias voltage applied between the tip and the surface to be examined allows electrons to tunnel through the vacuum separating them when the tip is brought very close to the surface to be examined. The resulting tunnelling current is a function of the sample's local density of states (LDOS), applied voltage, and tip location. The

current is monitored as the tip scans across the surface, and the data is generally displayed as an image.

Scanning tunnelling spectroscopy is a modification of the technique that involves holding the tip in a constant position above the surface while varying the bias voltage and documenting the resulting shift in current. The local density of electronic states can be reconstructed using this technique. To infer the properties and interactions of electrons in the studied material, this is often achieved in strong magnetic fields and in the presence of impurities.

Scanning tunnelling microscopy is a difficult technique to learn because it necessitates extremely clean and stable surfaces, sharp tips, strong vibration isolation, and sophisticated electronics.

2000S AND LATER

The environmental scanning electron microscope (ESEM) is a scanning electron microscope (SEM) that uses a gaseous atmosphere in the specimen chamber to capture electron micrographs of wet, uncoated, or both wet and uncoated specimens. While modified SEMs had previously been effective in viewing wet specimens in internal chambers, the ESEM's advanced electron detectors (rather than the regular Everhart-Thornley detector) and differential pumping systems, allow the electron beam to be transferred from the high vacuum in the gun region to the high pressure attainable in the specimen chamber, make it a complete device. Gerasimos Danilatos designed the instrument while a student at the University of New South Wales.

Dame Pratibha Gai invented the in-situ atomic-resolution environmental transmission electron microscope (ETEM). She decided not to patent her invention in order to further the advancement of science. Gai, who holds a chair in electron microscopy at the University of York and was co-director of the York JEOL Nanocentre, succeeded in developing a microscope capable of perceiving chemical reactions at the atomic scale in 2009. 